浙江大學
ZHEJIANG UNIVERSITY



Signal and System
Lab Report

Experiment Name	Lab03
Experiment Location	西 2-310
Name	Zhixuan Zheng
Student ID	3230103913
Experiment Date	April 3, 2025
Instructor	Xiaoming Chen

Contents

1 Problem1 **1**

1.1 Problem . . . 1

1.2 Answer . . . 1

1.2.1 Answer(a) . . . 1

1.2.2 Answer(b) . . . 1

1.2.3 Answer(c) . . . 2

2 Problem2 **3**

2.1 Problem . . . 3

2.2 Answer . . . 3

3 Problem3 a-g **4**

3.1 Problem . . . 4

3.2 Answer . . . 4

3.2.1 Answer a . . . 4

3.2.2 Answer b . . . 4

3.2.3 Answer c . . . 5

3.2.4 Answer d,e . . . 5

3.2.5 Answer f,g . . . 5

4 Problem3 h-p **7**

4.1 Problem . . . 7

4.2 Answer . . . 7

4.2.1 Answer h . . . 7

4.2.2 Answer i . . . 8

4.2.3 Answer j . . . 8

4.2.4 Answer k . . . 9

4.2.5 Answer l . . . 9

4.2.6 Answer m . . . 10

4.2.7 Answer n . . . 10

4.2.8 Answer o . . . 10

4.2.9 Answer p . . . 10

5 Code **11**

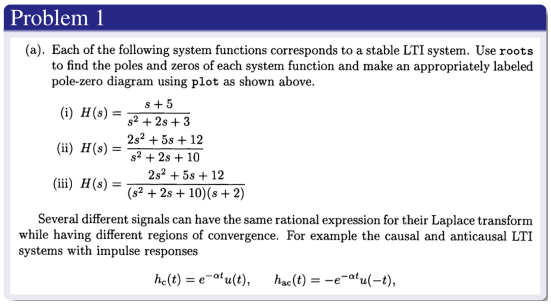
5.1 Code1 11

5.2 Code2 13

5.3 Code3 14

1 Problem1

1.1 Problem


Problem 1

(a). Each of the following system functions corresponds to a stable LTI system. Use `roots` to find the poles and zeros of each system function and make an appropriately labeled pole-zero diagram using `plot` as shown above.

(i) $H(s) = \frac{s+5}{s^2+2s+3}$

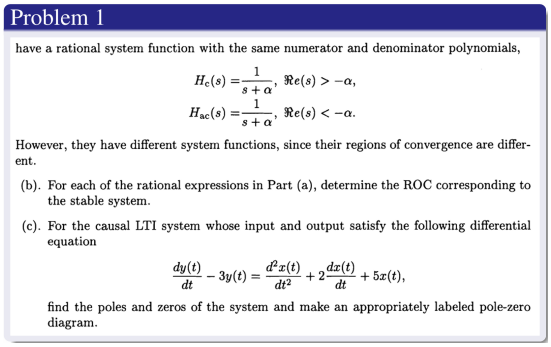
(ii) $H(s) = \frac{2s^2+5s+12}{s^2+2s+10}$

(iii) $H(s) = \frac{2s^2+5s+12}{(s^2+2s+10)(s+2)}$

Several different signals can have the same rational expression for their Laplace transform while having different regions of convergence. For example the causal and anticausal LTI systems with impulse responses

$$h_c(t) = e^{-\alpha t}u(t), \qquad h_{ac}(t) = -e^{-\alpha t}u(-t),$$

Figure 1: Problem1


Problem 1

have a rational system function with the same numerator and denominator polynomials,

$$H_c(s) = \frac{1}{s+\alpha}, \ \Re e(s) > -\alpha,$$
$$H_{ac}(s) = \frac{1}{s+\alpha}, \ \Re e(s) < -\alpha.$$

However, they have different system functions, since their regions of convergence are different.

(b). For each of the rational expressions in Part (a), determine the ROC corresponding to the stable system.

(c). For the causal LTI system whose input and output satisfy the following differential equation

$$\frac{dy(t)}{dt} - 3y(t) = \frac{d^2x(t)}{dt^2} + 2\frac{dx(t)}{dt} + 5x(t),$$

find the poles and zeros of the system and make an appropriately labeled pole-zero diagram.

Figure 2: Problem1

1.2 Answer

1.2.1 Answer(a)

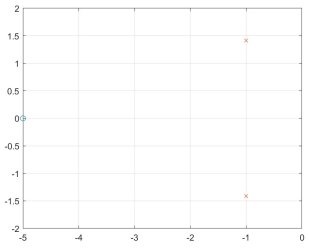


Figure 3: Answer1a (i)

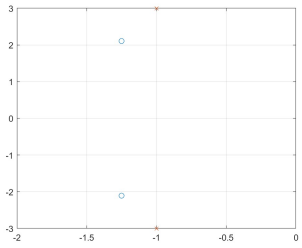


Figure 4: Answer1a (ii)

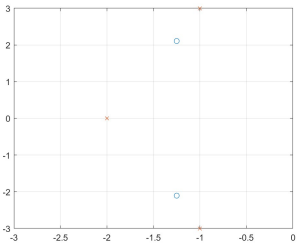


Figure 5: Answer1a (iii)

1.2.2 Answer(b)

(i)$ROC : Re(s) > -1$

(ii)$ROC : Re(s) > -1$

(iii)$ROC : Re(s) > -1$

1.2.3 Answer(c)

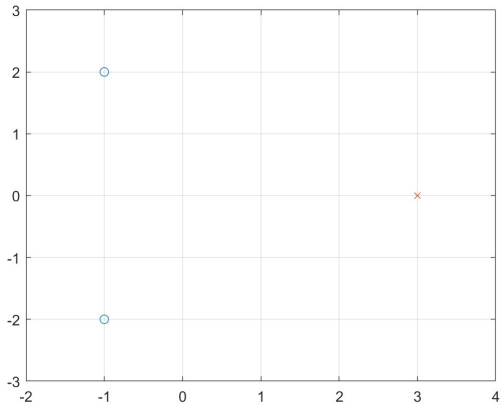


Figure 6: Answer1c

2 Problem2

2.1 Problem

Problem 2

For the exercises in this problem, you will need the M-file dpzplot.m. The function **dpzplot(b,a)** plots the poles and zeros of discrete-time systems. The inputs to **dpzplot** are in the same format as **filter**, and **dpzplot** will automatically append an appropriate number of zeros to a or to b if the numerator and denominator polynomials are not of the same order. Also, **dpzplot** will include the unit circle in the plot as well as an indication of the number of poles or zeros at the origin-if there are more than one.

Figure 7: Problem2

Problem 2

- (a). Use **dpzplot** to plot the poles and zeros for $H(z)$ in Eq. (1).
- (b). Use **dpzplot** to plot the poles and zeros for a filter which satisfies the difference equation

$$y[n] + y[n-1] + 0.5y[n-2] = x[n]. \tag{3}$$

- (c). Use **dpzplot** to plot the poles and zeros for a filter which satisfies the difference equation

$$y[n]-1.25y[n-1]+0.75y[n-2]-0.125y[n-3] = x[n]+0.5x[n-1]. \tag{4}$$

Figure 8: Problem2

2.2 Answer

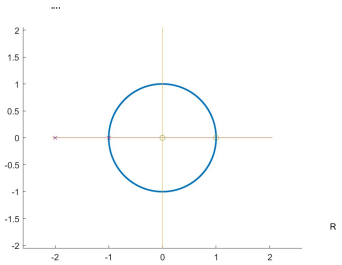


Figure 9: Answer2a

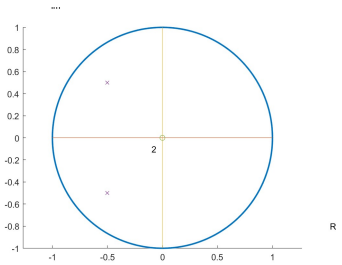


Figure 10: Answer2b

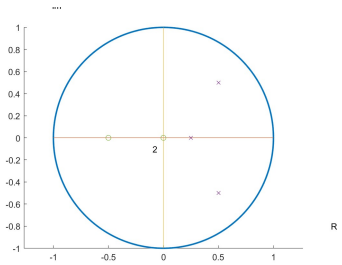
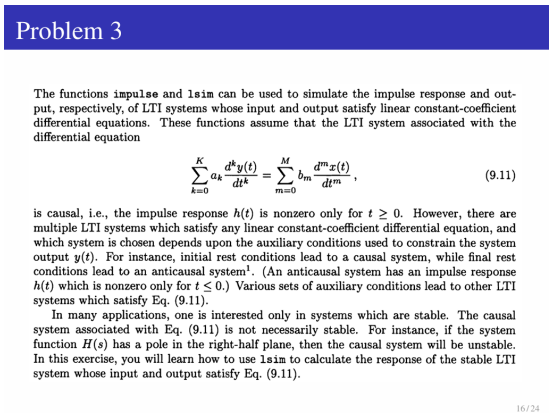


Figure 11: Answer2c

3 Problem3 a-g

3.1 Problem


Problem 3

The functions impulse and lsim can be used to simulate the impulse response and output, respectively, of LTI systems whose input and output satisfy linear constant-coefficient differential equations. These functions assume that the LTI system associated with the differential equation

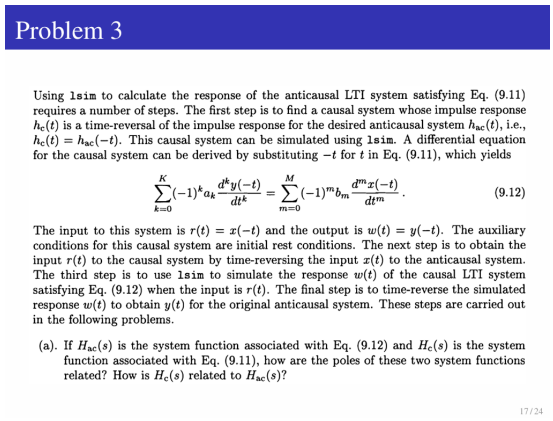
$$\sum_{k=0}^{K} a_k \frac{d^k y(t)}{dt^k} = \sum_{m=0}^{M} b_m \frac{d^m x(t)}{dt^m}, \tag{9.11}$$

is causal, i.e., the impulse response $h(t)$ is nonzero only for $t \geq 0$. However, there are multiple LTI systems which satisfy any linear constant-coefficient differential equation, and which system is chosen depends upon the auxiliary conditions used to constrain the system output $y(t)$. For instance, initial rest conditions lead to a causal system, while final rest conditions lead to an anticausal system[1]. (An anticausal system has an impulse response $h(t)$ which is nonzero only for $t \leq 0$.) Various sets of auxiliary conditions lead to other LTI systems which satisfy Eq. (9.11).

In many applications, one is interested only in systems which are stable. The causal system associated with Eq. (9.11) is not necessarily stable. For instance, if the system function $H(s)$ has a pole in the right-half plane, then the causal system will be unstable. In this exercise, you will learn how to use lsim to calculate the response of the stable LTI system whose input and output satisfy Eq. (9.11).

16 / 24

Figure 12: Problem3


Problem 3

Using lsim to calculate the response of the anticausal LTI system satisfying Eq. (9.11) requires a number of steps. The first step is to find a causal system whose impulse response $h_c(t)$ is a time-reversal of the impulse response for the desired anticausal system $h_{ac}(t)$, i.e., $h_c(t) = h_{ac}(-t)$. This causal system can be simulated using lsim. A differential equation for the causal system can be derived by substituting $-t$ for t in Eq. (9.11), which yields

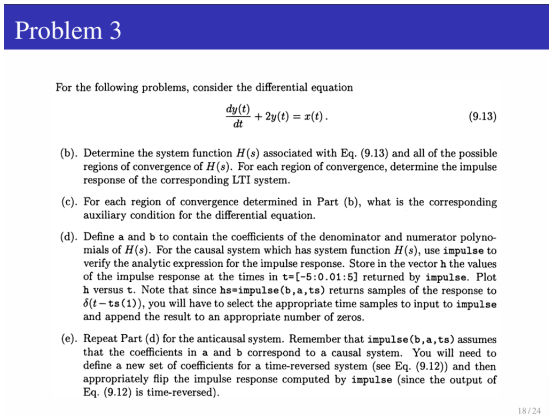
$$\sum_{k=0}^{K} (-1)^k a_k \frac{d^k y(-t)}{dt^k} = \sum_{m=0}^{M} (-1)^m b_m \frac{d^m x(-t)}{dt^m}. \tag{9.12}$$

The input to this system is $r(t) = x(-t)$ and the output is $w(t) = y(-t)$. The auxiliary conditions for this causal system are initial rest conditions. The next step is to obtain the input $r(t)$ to the causal system by time-reversing the input $x(t)$ to the anticausal system. The third step is to use lsim to simulate the response $w(t)$ of the causal LTI system satisfying Eq. (9.12) when the input is $r(t)$. The final step is to time-reverse the simulated response $w(t)$ to obtain $y(t)$ for the original anticausal system. These steps are carried out in the following problems.

(a). If $H_{ac}(s)$ is the system function associated with Eq. (9.12) and $H_c(s)$ is the system function associated with Eq. (9.11), how are the poles of these two system functions related? How is $H_c(s)$ related to $H_{ac}(s)$?

17 / 24

Figure 13: Problem3


Problem 3

For the following problems, consider the differential equation

$$\frac{dy(t)}{dt} + 2y(t) = x(t). \tag{9.13}$$

(b). Determine the system function $H(s)$ associated with Eq. (9.13) and all of the possible regions of convergence of $H(s)$. For each region of convergence, determine the impulse response of the corresponding LTI system.

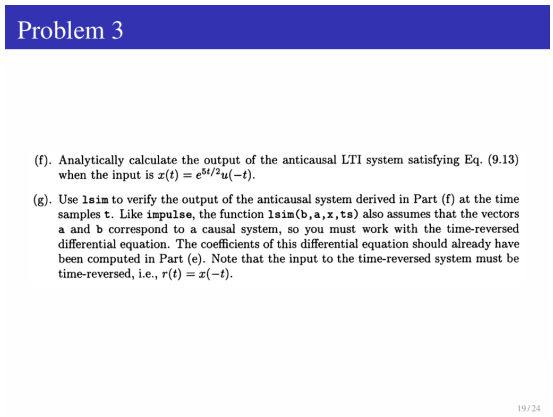
(c). For each region of convergence determined in Part (b), what is the corresponding auxiliary condition for the differential equation.

(d). Define a and b to contain the coefficients of the denominator and numerator polynomials of $H(s)$. For the causal system which has system function $H(s)$, use impulse to verify the analytic expression for the impulse response. Store in the vector h the values of the impulse response at the times in t=[-5:0.01:5] returned by impulse. Plot h versus t. Note that since hs=impulse(b,a,ts) returns samples of the response to $\delta(t - \texttt{ts(1)})$, you will have to select the appropriate time samples to input to impulse and append the result to an appropriate number of zeros.

(e). Repeat Part (d) for the anticausal system. Remember that impulse(b,a,ts) assumes that the coefficients in a and b correspond to a causal system. You will need to define a new set of coefficients for a time-reversed system (see Eq. (9.12)) and then appropriately flip the impulse response computed by impulse (since the output of Eq. (9.12) is time-reversed).

18 / 24

Figure 14: Problem3


Problem 3

(f). Analytically calculate the output of the anticausal LTI system satisfying Eq. (9.13) when the input is $x(t) = e^{5t/2}u(-t)$.

(g). Use lsim to verify the output of the anticausal system derived in Part (f) at the time samples t. Like impulse, the function lsim(b,a,x,ts) also assumes that the vectors a and b correspond to a causal system, so you must work with the time-reversed differential equation. The coefficients of this differential equation should already have been computed in Part (e). Note that the input to the time-reversed system must be time-reversed, i.e., $r(t) = x(-t)$.

19 / 24

Figure 15: Problem3

3.2 Answer

3.2.1 Answer a

if s_0 the pole of $H_c(s)$,then $-s_0$ is the pole of $H_{ac}(s)$

$$H_{ac}(s) = H_c(-s)$$

3.2.2 Answer b

$$H(s) = \frac{Y(s)}{X(s)} = \frac{1}{s+2}$$

$$\text{ROC: Re}\{s\} > -2 \; h(t) = e^{-2t}u(t)$$

$$\text{ROC: Re}\{s\} < -2 \; h(t) = -e^{-2t}u(-t)$$

3.2.3 Answer c

ROC: Re$\{s\} > -2$ initial rest conditions

ROC: Re$\{s\} < -2$ final rest conditions

3.2.4 Answer d,e

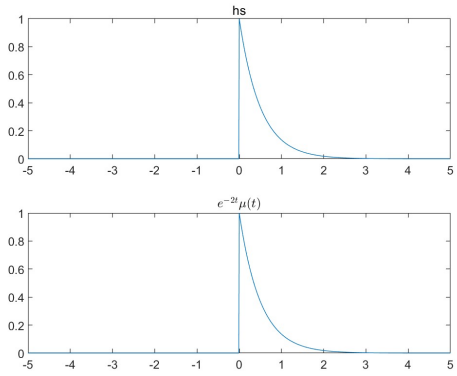


Figure 16: Answer3d

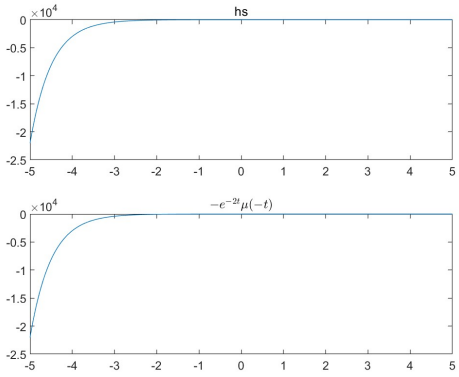


Figure 17: Answer3e

3.2.5 Answer f,g

$$X(s) = -\frac{1}{s - 2.5}$$

$$Y(s) = H(s)X(s) = -\frac{1}{(s+2)(s-2.5)}$$

$$\Rightarrow y = (\frac{2}{9} \cdot e^{\frac{5}{2}t} - \frac{2}{9} \cdot e^{-2t}) \cdot u(-t)$$

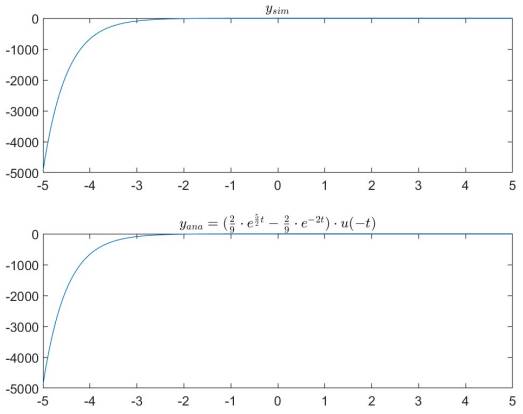


Figure 18: Answer3g

4 Problem3 h-p

4.1 Problem

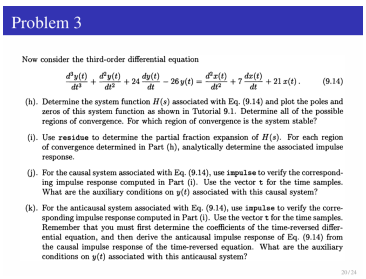


Figure 19: Problem3

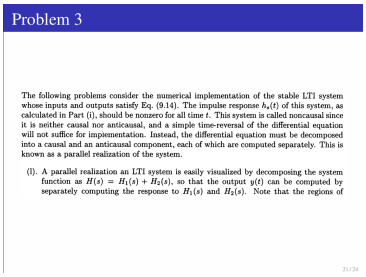


Figure 20: Problem3

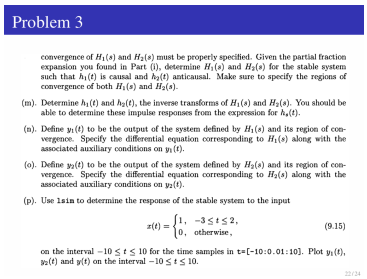


Figure 21: Problem3

4.2 Answer

4.2.1 Answer h

$$\left(s^3 + s^2 + 24s - 26\right) Y(s) = \left(s^2 + 7s + 21\right) X(s)$$

$$H(s) = \frac{Y(s)}{X(s)} = \frac{s^2 + 7s + 21}{s^3 + s^2 + 24s - 26}$$

(1). Right-sided (Causal): $\Re(s) > 1$ (excludes the imaginary axis, unstable).

(2). Two-sided: $-1 < \Re(s) < 1$ (includes the imaginary axis, stable).

(3). Left-sided (Anticausal): $\Re(s) < -1$ (excludes the imaginary axis, unstable).

The system is stable only for the ROC $-1 < \Re(s) < 1$, as it includes the $j\omega$-axis.

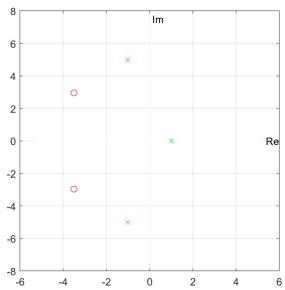


Figure 22: Answer3h

4.2.2 Answer i

l the values of [r,p,k] are demonstrated below

```
r =

   0.0000 - 0.5000i
   0.0000 + 0.5000i
   1.0000 + 0.0000i


p =

  -1.0000 + 5.0000i
  -1.0000 - 5.0000i
   1.0000 + 0.0000i


k =

     []
```

Figure 23: Answer3i

$$H(s) = \frac{1}{s-1} + \frac{-0.5j}{s-(-1+5j)} + \frac{0.5j}{s-(-1-5j)}$$

$$1.\,\mathrm{ROC} : \Re(s) > 1, \quad h(t) = \left[e^{t} + e^{-t}\sin(5t)\right] u(t)$$

$$2.\,\mathrm{ROC} : \Re(s) < -1, \quad h(t) = -\left[e^{t} + e^{-t}\sin(5t)\right] u(-t)$$

$$3.\,\mathrm{ROC} : -1 < \Re(s) < 1, \quad h(t) = -e^{t}u(-t) + e^{-t}\sin(5t)u(t)$$

4.2.3 Answer j

$$case1 : h(t) = \left[e^{t} + e^{-t}\sin(5t)\right] u(t)$$

The auxiliary conditions are initial rest conditions

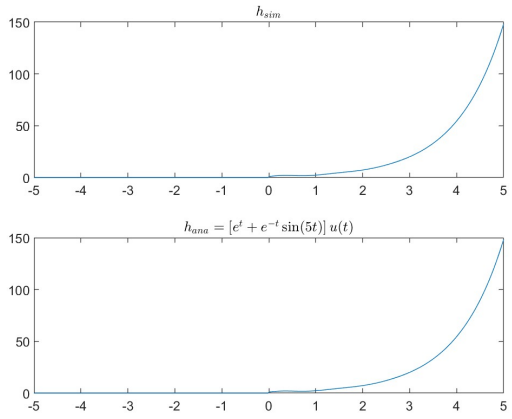


Figure 24: Answer3j

4.2.4 Answer k

$$case2 : h(t) = -\left[e^{t} + e^{-t}\sin(5t)\right] u(-t)$$

The auxiliary conditions are final rest conditions

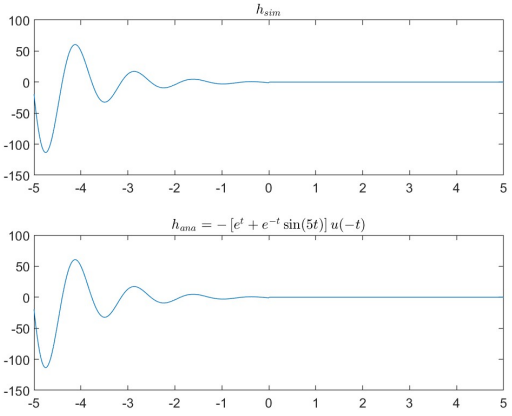


Figure 25: Answer3k

4.2.5 Answer l

$$case3 : h(t) = -e^{t}u(-t) + e^{-t}\sin(5t)u(t)$$

$$H(s) = \underbrace{\frac{5}{(s+1)^2 + 25}}_{H_1(s)} + \underbrace{\frac{1}{s-1}}_{H_2(s)}$$

$$H_1(s) = \frac{5}{(s+1)^2+25} \quad \text{with ROC: } \Re(s) > -1$$

$$H_2(s) = \frac{1}{s-1} \quad \text{with ROC: } \Re(s) < 1$$

4.2.6 Answer m

$$h(t) = \underbrace{e^{-t}\sin(5t)u(t)}_{h_1(t)} + \underbrace{[-e^{t}u(-t)]}_{h_2(t)}$$

4.2.7 Answer n

$$\frac{d^2y(t)}{dt^2} + 2\frac{dy(t)}{dt} + 26y(t) = 5x(t)$$

The auxiliary conditions are initial rest conditions

4.2.8 Answer o

$$\frac{dy(t)}{dt} - y(t) = x(t)$$

The auxiliary conditions are final rest conditions

4.2.9 Answer p

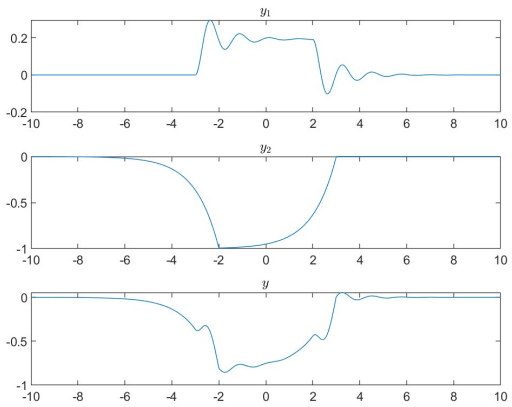


Figure 26: Answer3p

5 Code

5.1 Code1

Listing 1: Answer1a

```
clear;
clc;

b1 = [1 5];
b2 = [2 5 12];
b3 = [2 5 12];

zs1 = roots(b1);
zs2 = roots(b2);
zs3 = roots(b3);

a1 = [1 2 3];
a2 = [1 2 10];
a31 = [1 2 10];
a32 = [1 2];
a3 = conv(a31,a32);

ps1 = roots(a1);
ps2 = roots(a2);
ps3 = roots(a3);

figure
plot(real(zs1),imag(zs1),'o');
hold on;
plot(real(ps1),imag(ps1),'x');
grid;
axis([-5 0 -2 2]);

figure
plot(real(zs2),imag(zs2),'o');
hold on;
plot(real(ps2),imag(ps2),'x');
grid;
```

```
34  axis([-2 0 -3 3]);
35
36  figure
37  plot(real(zs3),imag(zs3),'o');
38  hold on;
39  plot(real(ps3),imag(ps3),'x');
40  grid;
41  axis([-3 0 -3 3]);
```

Listing 2: Answer1c

```
1   clear;
2   clc;
3
4   b = [1 2 5];
5   zs = roots(b);
6
7   a = [1 -3];
8   ps = roots(a);
9
10  plot(real(zs),imag(zs),'o');
11  hold on;
12  plot(real(ps),imag(ps),'x');
13  grid;
14  axis([-2 4 -3 3]);
```

5.2 Code2

Listing 3: Answer2a

```
clear;
clc;

a=[1 3 2];
b=[1 -1 0];
dpzplot(b,a);
```

Listing 4: Answer2b

```
clear;
clc;

a=[1 3 2];
b=[1 -1 0];
dpzplot(b,a);
```

Listing 5: Answer2c

```
clear;
clc;

a=[1 -1.25 0.75 -0.125];
b=[1 0.5];
dpzplot(b,a);
```

5.3 Code3

Listing 6: Answer3d

```
clear;
clc;

t=-5:0.01:5;
t_=0:0.01:5;

a=[1 2];
b=1;

hs=impulse(b,a,t_);
hs=[zeros(length(t)-length(hs),1); hs];

y=exp(-2*t).*(t>=0);

subplot(2,1,1);
plot(t,hs);
title('hs');

subplot(2,1,2);
plot(t,y);
title('$e^{-2t}\mu (t)$','Interpreter','latex')
```

Listing 7: Answer3e

```
clear;
clc;

t=-5:0.01:5;
t_=0:0.01:5;

a=[-1 2];
b=1;

hs=impulse(b,a,t_);
hs=[zeros(length(t)-length(hs),1);hs];
hs=flip(hs);
```

```
13
14  y=-exp(-2*t).*(t<=0);
15
16  subplot(2,1,1);
17  plot(t,hs);
18  title('hs');
19
20  subplot(2,1,2);
21  plot(t,y);
22  title('$-e^{-2t}\mu (-t)$','Interpreter','latex')
```

Listing 8: Answer3f

```
1   clear;
2   clc;
3
4   t=-5:0.01:5;
5   t_=0:0.01:5;
6
7   a=[-1 2];
8   b=1;
9
10  x=exp(5*t/2).*(t<=0);
11  x_r=exp(-5*t/2).*(t>=0);
12
13  y_sim=lsim(b,a,x_r,t);
14  y_sim=flip(y_sim);
15
16  y_ana = (2/9)*exp(2.5*t).*(t <= 0) - (2/9)*exp(-2*t).*(t <= 0);
17
18  subplot(2,1,1);
19  plot(t,y_sim);
20  title('$y_{sim}$','Interpreter','latex')
21
22  subplot(2,1,2);
23  plot(t,y_ana);
24  title('$y_{ana}=(\frac{2}{9} \cdot e^{\frac{5}{2}t} - \frac{2}{9} \
      cdot e^{-2t}) \cdot u(-t) $','Interpreter','latex')
```

Listing 9: Answer3h

```
clear;
clc;

a=[1 1 24 -26];
b=[1 7 21];
pzplot(b,a);
```

Listing 10: Answer3i

```
clear;
clc;

b=[1 7 21];
a=[1 1 24 -26];

[r,p,k]=residue(b,a);
```

Listing 11: Answer3j

```
clear;
clc;

a = [1 1 24 -26];
b = [1 7 21];
t = -5:0.01:5;
t_ = 0:0.01:5;

h_ana = (exp(t)+exp(-t).*sin(5*t)).*(t>=0);

h_sim = impulse(b,a,t_);
h_sim = [zeros(500,1);h_sim];

subplot(2,1,1);
plot(t,h_sim);
title('$h_{sim}$','Interpreter','latex')

subplot(2,1,2);
plot(t,h_ana);
```

```
20  title('$h_{ana}=\left[ e^t + e^{-t} \sin(5t) \right] u(t) $','
        Interpreter','latex')
```

Listing 12: Answer3k

```
1   clear;
2   clc;
3
4   a = [-1 1 -24 -26];
5   b = [1 -7 21];
6   t = -5:0.01:5;
7   t_ = 0:0.01:5;
8
9   h_ana = -(exp(t)+exp(-t).*sin(5*t)).*(t<=0);
10
11  h_sim = impulse(b,a,t_);
12  h_sim = [zeros(500,1);h_sim];
13  h_sim = flip(h_sim);
14
15  subplot(2,1,1);
16  plot(t,h_sim);
17  title('$h_{sim}$','Interpreter','latex')
18
19  subplot(2,1,2);
20  plot(t,h_ana);
21  title('$h_{ana}=-\left[e^t + e^{-t} \sin(5t) \right] u(-t) $','
        Interpreter','latex')
```

Listing 13: Answer3p

```
1   clear;
2   clc;
3
4   t=-10:0.01:10;
5
6   a1=[1 2 26];
7   b1=5;
8
9   x=(t>=-3).*(t<=2);
10  x_r=(t>=-2).*(t<=3);
```

```
11
12  y1=lsim(b1,a1,x,t);
13
14  a2=[-1 -1];
15  b2=1;
16
17  y2=lsim(b2,a2,x_r,t);
18  y2=flip(y2);
19
20  subplot(3,1,1);
21  plot(t,y1);
22  title('$y_{1}$','Interpreter','latex')
23
24  subplot(3,1,2);
25  plot(t,y2);
26  title('$y_{2}$','Interpreter','latex')
27
28  subplot(3,1,3);
29  plot(t,y1+y2);
30  title('$y$','Interpreter','latex')
```